

Boronic ionogel electrolytes to improve lithium transport for Li-ion batteries



Albert S. Lee^{a,1}, Jin Hong Lee^{a,b,1}, Soon Man Hong^{a,c}, Jong-Chan Lee^b,
Seung Sang Hwang^{a,c}, Chong Min Koo^{a,c,*}

^a Materials Architecturing Research Center, Korea Institute of Science and Technology, 5 Hwarangno 14-gil, Seongbuk-gu, Seoul 02792, Republic of Korea

^b School of Chemical and Biological Engineering and Institute of Chemical Processes, Seoul National University, 599 Gwanak-ro, Gwanak-gu, Seoul 151-742, Republic of Korea

^c Nanomaterials Science and Engineering, University of Science and Technology, 217 Gajungro, Yuseong-Gu, Daejeon 34113, Republic of Korea

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ABSTRACT

Boron containing ionogels were fabricated through chemical crosslinking of boron allyloxide with polyethylene glycol dimethacrylate in an ionic liquid electrolyte solution to obtain mechanically robust gels. Because of the relatively small concentration of crosslinking agent required to fully solidify the ionic liquid electrolyte, good characters of high ionic conductivity, high thermal stability, and good electrochemical stability were observed. A spectroscopic investigation of the boronic ionogels revealed that the lithium mobility was noticeably enhanced compared with ionogels fabricated without the boronic crosslinker, leading to promising Li-ion battery performance at elevated temperatures.

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1. Introduction

Lithium-ion batteries are highly sought out as the energy storage devices for various applications, ranging from portable electronic devices to hybrid vehicles [1]. As these electronic devices have become more common in everyday human life, battery safety has become one of the key issues to be required along with their high performance [2,3].

The need to fortify the safety of Li-ion batteries stems from the thermal instability of conventional commercial non-aqueous liquid electrolytes. In these systems, the volatile organic solvents under high temperature conditions render volumetric expansion of cells and electrolyte leakage, which in turn lead to cell short circuit and possible explosions [4,5]. As such, the use of ionic liquids and their solidified gel counterparts, so-called ‘ionogels’, have received much attention in recent years [6–8].

Ionogels can be fabricated through one of the methods that includes swelling of ionic liquid in physically crosslinked polymers [9] or in chemically crosslinked polymers with multifunctional crosslinkable organic moieties through either UV radiation or

thermal treatment [10]. While both methods have proved to be useful in various cases, the *in-situ* mix-and-cure system of using multifunctional crosslinkers can be considered the most practical method for device fabrication. To date, a vast amount of literature studies have been devoted to study the various organic [10], hybrid [11,12], and inorganic [13] materials as ionic liquid crosslinkers for the fabrication of ionogels.

While the safety of Li-ion batteries can be improved through fabrication of non-leaking ionogels, the inevitable depreciation in ionic conductivity is caused by the sluggish mass transport of ionogels synthesized with large crosslinker concentrations. Moreover, ionic liquid electrolytes, though having inherently high ionic conductivity, are composed of lithium salts dissolved in quaternary ammonium-based ionic liquids [14], leading to the well-known phenomenon of two-cation competition between the ionic liquid cation and lithium cation [15]. As such, the strategy of improvement in lithium mobility [12] within the solidified ionogels matrices to enhance Li-ion battery performance is still an ongoing challenge.

Because of these limitations, improvement of lithium transport in ionic liquids and their solidified ionogels counterparts have been studied by many research groups. One of the most investigated methods for this goal is the use of non-ion conducting moieties to effectively trap the anions of both the lithium salt and the ionic liquid to enhance the relative mobility of lithium ions. The most

* Corresponding author.

E-mail address: koo@kist.re.kr (C.M. Koo).

¹ These authors contributed equally to this work.

representative example of this is the use of boron-containing functionalities. Boron, with an empty d-orbital, is highly electro-positive when in its trivalent neutral state [16], rendering a partial positive charge, which functions to trap anions in ionic solutions. Previous reports have detailed complicated synthetic procedures and fabrication methods to develop boron-containing salts [17], boron-containing polymers [18,19], and boron-containing composites [20] for the enhancement of lithium transfer or lithium mobility. However, most of these studies have entailed the use of chelated boron anions [17,19,20] that are either salts or ionic polymers without the appropriate thermal or ionic conductivity requirements for practical applications.

In this study, we investigated a more practical methodology for the enhancement of lithium mobility via application of a highly soluble and crosslinkable boron oxide precursor as gelating agent to solidify an ionic liquid electrolyte solution to yield boron containing ionogel electrolytes. Spectroscopic analyses of these new boronic ionogel electrolytes revealed that lithium mobility is highly enhanced even compared with the neat liquid-state ionic liquid, leading to highly promising Li-ion battery performance.

2. Experimental

2.1. Materials

Boron allyloxide (BAO) (Gelest, 99%) and polyethylene glycol dimethacrylate (PEGDMA) (Sigma Aldrich, M_w = 500, 99%) were purified over an inhibitor removal column (Sigma Aldrich). *N*-butyl-*N*-methylpyrrolidinium bis(trifluoromethylsulfonyl)imide (BMPTFSI) (>99.9%, battery grade) was purchased from CTRL, Korea. Photoinitiator Irgacure 184 (BASF), lithium bis(trifluoromethylsulfonyl)imide (LiTFSI) (Aldrich, 99.9%, battery grade) and all solvents were used as received.

2.2. Characterization

Fourier transform infrared spectroscopy examinations were conducted with a Perkin-Elmer FTIR system Spectrum-GX. Raman spectra were obtained with a Renishaw InVia spectrometer equipped with 633 nm HeNe laser. Thermal gravimetric analysis was carried out with TA instrument (TGA 2950) at a heating rate of $10^\circ\text{C min}^{-1}$ under N_2 atmosphere. Rheological properties were measured using a MCR 302 rheometer (Anton Paar, Austria) instrument with parallel plate geometry (25 mm diameter). All rheological examinations were performed in the linear viscoelastic region under nitrogen atmosphere. Solid-state ^7Li NMR spectra were obtained on a Varian Unity (^1H : 400 MHz, ^7Li : 155.45 MHz) with spinning frequency held constant at 10 kHz with a pulse delay time of 4 s. Samples were packed into 7.5 mm zirconia rotors and sealed with Kel-F short caps.

The ionic conductivity values were determined using a complex impedance analyzer (Bio-Logics, VMP3) over frequencies ranging from 1 Hz to 1 MHz at AC amplitude of 10 mV. The anodic stability

of the ionogel electrolytes was evaluated using a linear sweep voltammetry technique in a beaker cell at a scan rate of 1.0 mV s^{-1} . A platinum working electrode with 2 mm diameter was used with lithium metal as both the counter and reference electrode. Electrochemical measurements of the ionogel electrolyte were conducted using CR-2032 coin cells consisting of a separator, Li metal, and a LiFePO_4 cathode (90 wt % LiFePO_4 , 5 wt % carbon black, 5 wt % PVdF). LiFePO_4 (Life Power P2, Clarinat) was used as received. All the cells were assembled in argon-charged glove box. After fabrication, the cells with pre-gel solution were subjected to UV cross-linking under total UV irradiation of 3 J cm^{-2} with a Hitachi UV Spot Cure System apparatus. The galvanostatic charge-discharge experiments were carried out in a voltage range of 2.5–4.2 V using a battery cycler (Wonatech, WBCS3000) at 50°C .

2.3. Fabrication of boronic ionogels

The fabrication of boronic ionogels (BI) began with the dissolution of BAO in PEGDMA at 1:1 weight ratio. This homogeneous liquid mixture, along with photoinitiator Irgacure 184 (1 wt % with respect to the crosslinker or mixed crosslinker solution) was then added to a 1 M LiTFSI BMPTFSI ionic liquid electrolyte solution at 10 wt % and stirred vigorously at room temperature. The resulting pre-gel solution was UV-cured under mild conditions to yield homogeneous ionogels, named as BI-10. For comparison, ionogels without the boron allyloxide crosslinker with only 10 wt % PEGDMA were also fabricated for comparison, named as PEGI-10.

3. Results and discussion

3.1. Fabrication and characterization of boronic ionogel electrolytes

As shown in Fig. 1, the ionogels were fabricated via UV-induced curing process accomplished with chemical crosslinking reaction of methacryl- and allyl- groups of (PEGDMA and BAO, respectively, within the ionic liquid solution of 1 M LiTFSI in BMPTFSI. After UV irradiation, the ionic liquid solution was entrapped within the crosslinked polymeric network structure, resulting in a homogeneous and mechanically stable ionogel. The FTIR spectra changes (Fig. 2a) showed that $\text{C}=\text{C}$ bonds at 1637 cm^{-1} completely disappeared, demonstrating that the crosslinking reaction was completed. As shown in Fig. 2b photograph, the ionic liquid solution was a light yellow transparent liquid, whereas the ionogel exhibited non-flowing gel-state albeit the vial containing ionogel was tilted vertically upside down.

The solid-like gel properties of prepared samples were confirmed through examination of their rheological behaviors before and after curing process. The changes of dynamic viscoelastic properties of neat ionic liquid and prepared ionogels as a function of the angular frequency are exhibited in Fig. 3. Before curing, the dynamic mechanical response of the neat ionic liquid solution revealed a liquid-like terminal behavior. The smaller

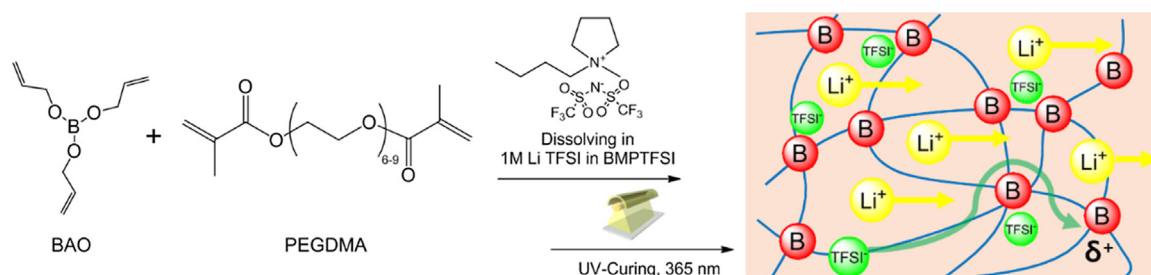


Fig. 1. Schematic diagram of fabrication for boronic ionogels.

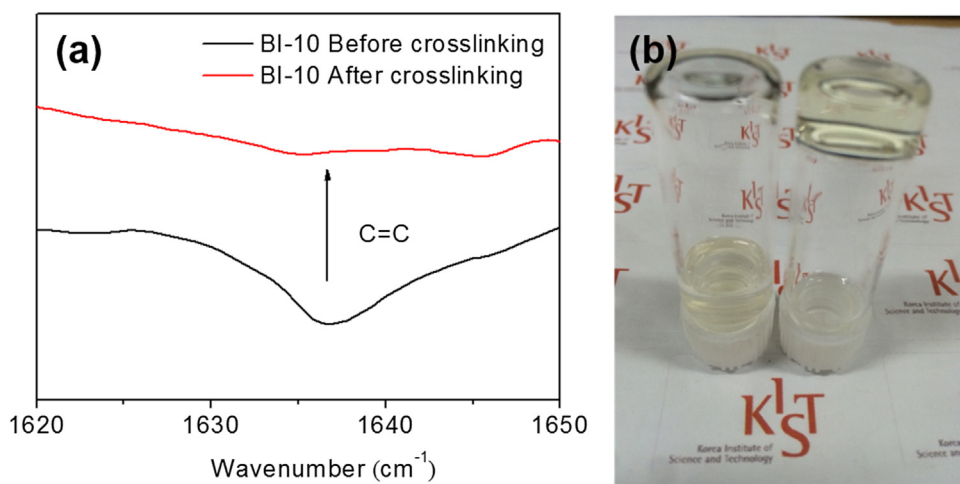


Fig. 2. (a) FTIR spectra and (b) optical photograph of BI-10 before (left) and after (right) crosslinking.

values of storage modulus G' compared with those of the loss modulus G'' were observed. Both dynamic moduli values exhibited power law dependency on frequency, a typical rheological behavior of viscous liquids [21]. On the other hand, for the given ionogel, G' was dominant over G'' , an indicative of a predominantly solid-like material response due to the formation of well-developed network structures. Moreover, both moduli were relatively independent over the measured frequency range, demonstrating that the obtained gelled structure was invariant to changes in the time period of the experiment.

3.2. Electrochemical and thermal stability

The thermal stabilities of ionogels were further examined with thermogravimetric analysis (TGA). As presented in Fig. 4a, no abrupt weight change in TGA trace was observed below 200 °C, an indication of the ionogel sufficient thermal stability for use as an electrolyte in lithium batteries at elevated temperatures. Moreover, the electrochemical stability of fabricated ionogels was investigated using linear sweep voltammetry (LSV) experiment. As shown in Fig. 4b, in the anodic region, there was no abrupt rise in current density associated with the oxidative decomposition of the electrolyte until the applied potential swept to about 5.0 V vs Li/Li⁺, demonstrating that these ionogels were electrochemically stable in the operation voltage range of 4 V class cathode materials.

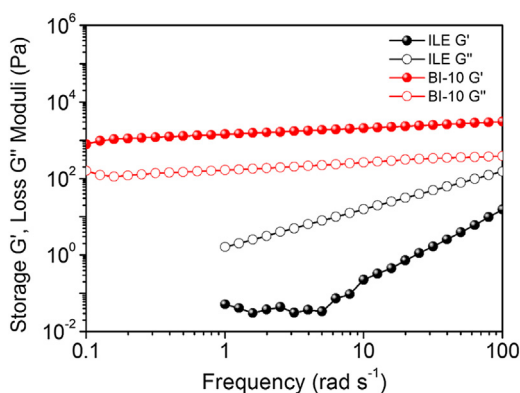


Fig. 3. Rheological properties of the ionic liquid electrolyte and BI-10.

3.3. Ion transport behavior in boronic ionogel electrolytes

The electrochemical properties of ionogels were investigated to see whether the incorporation of boron moieties into the network structure affects the ion transport behavior of ionogels. We first compared the measured ionic conductivity values as a function of temperature. The data for PEGI-10 sample and the ionic liquid electrolyte (ILE) were also investigated for comparison. As presented in Fig. 5, the ionogels fabricated via curing process provided good dimensional stability, while sacrificing ionic conductivity. This result is attributed to the formation of a network structure that retards the movement of ions. Furthermore, the incorporation of crosslinking agent dilutes the concentration of ions in the gel electrolytes when compared with the ionic liquid

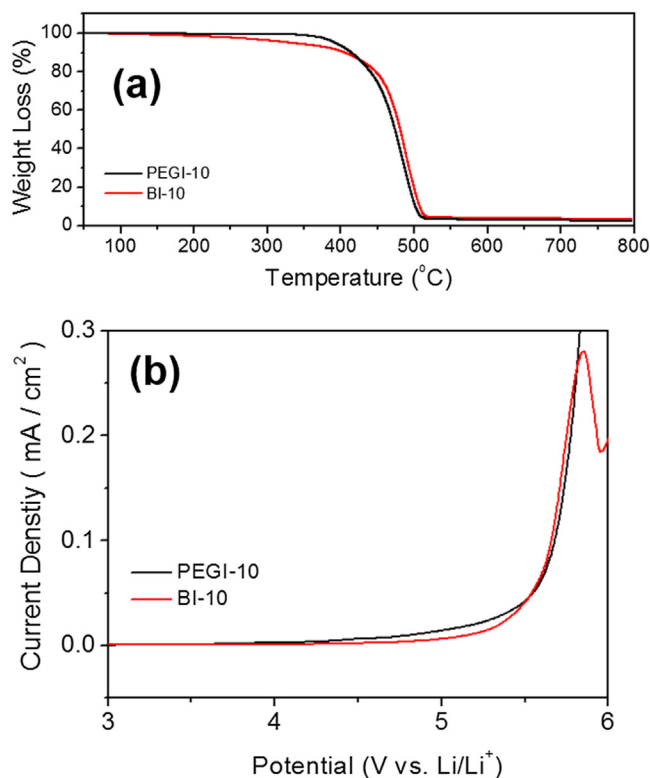


Fig. 4. (a) TGA thermograms and (b) Linear sweep voltammograms of PEGI-10 and BI-10 materials.

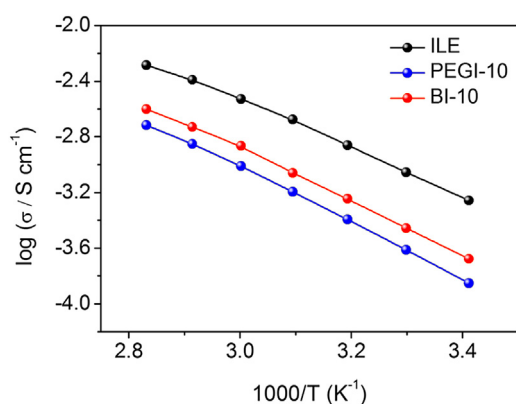


Fig. 5. Temperature dependency of ionic conductivities of ILE, PEGI-10, and BI-10 materials.

solution. However, the ionogel still provided a large ionic conductivity value in the order of $10^{-4} \text{ S cm}^{-1}$ at room temperature. The disparity in ionic conductivity value between both the electrolytes decreased continuously as the temperature was raised. Meanwhile, all electrolytes exhibited nearly Arrhenius response in the whole temperature range, suggesting that the ion transport in electrolyte systems was predominantly governed by the mobility of free ions [15]. Interestingly, the ionogel with boron moieties (BI-10) revealed a larger ionic conductivity value than one without boron moieties (PEGI), indicating that the ionic conductivity of the electrolytes are also dependent on their network structures. It should be noted that we fabricated both gel electrolytes with equal concentrations of cross-linker (10 wt %) to rule out a contributing factor for the decrease in ion concentrations. These measured ionic conductivities were comparable or higher than previous reports of chemically crosslinked ionogels. [12,13]

Additionally, we performed Raman spectroscopy experiments to compare the neat ionic liquid solution and ionogels regarding the extent of interaction between mobile ions in electrolyte systems (Fig. 6a). It has been observed that an intense Raman-active band at a frequency of 748 cm^{-1} is assigned for Li^+ coordinated TFSI^- ion pairs and a band at 742 cm^{-1} corresponds to free or dissociated TFSI^- ions [22,23]. As presented, the incorporation of crosslinking agent into the electrolyte led to changes in lithium cation coordination. Also noteworthy is that the

concentration of free ions for ionogel containing boron moieties was observed to be greater than that of ionogel without boron moieties, suggesting that boron moieties could effectively increase the concentration of dissociated mobile ions possibly due to the interaction between TFSI^- anion and boron moieties [19]. Generally, the ionic conductivity (σ) of an electrolyte may be given as the below equation

$$\sigma(T) = \sum n \times q \times \mu$$

where n is the number of the charge carriers, q is the charge on the charge carrier, and μ is the mobility of charge carriers [12]. The obtained ionic conductivity value was definitely influenced by both the number and mobility of ions. In this respect, the anion-trapping interaction could cause a decrease in TFSI^- ion mobility, leading to decrease in conductivity value. However, the ionogel with boron moieties exhibited larger ionic conductivity than the ionogel without boron moieties over a wide range of temperatures. We speculated that the increase in number of dissociated ions was able to positively compensate the impaired TFSI^- anion mobility, allowing for increase in ionic conductivity over their counterparts.

To better understand the observed ion transport behavior, we investigated the relative mobility of lithium ions in the electrolytes via solid-state ^7Li NMR examination, as narrower linewidths reflect a change in local environments of the lithium cations [12]. As shown in Fig. 6b, the sharp and singular peak in the neat ionic liquid electrolyte spectrum was an indication of high lithium mobility due to its liquid state. For the PEGI-10 ionogels, the ^7Li NMR linewidths were larger than those in the neat ionic liquid electrolyte as a result of the limiting mass transport through the gel state compared with that in the liquid state ionic liquid electrolyte. However, the degree to which the line broadening was pronouncedly different, as the ^7Li NMR linewidth for the BI-10 ionogel was close to that of the ionic liquid electrolyte, was suggesting that the ionogel with boron moieties contains a larger concentration of mobile lithium ions because of the anion-trapping ability of boron moieties. The observation was in good agreement with the information from the Raman spectroscopy results. This point was confirmed further by observation of the temperature dependency of solid state ^7Li NMR spectra (Fig. 6c–e). As the temperature increased above $\sim 35^\circ\text{C}$, a stark sharpening effect of the Li cations was observed, which correlated well to the melt-phase state of the ionic liquid electrolyte confined in the crosslinked matrix. Therefore, it is reasonable to believe that

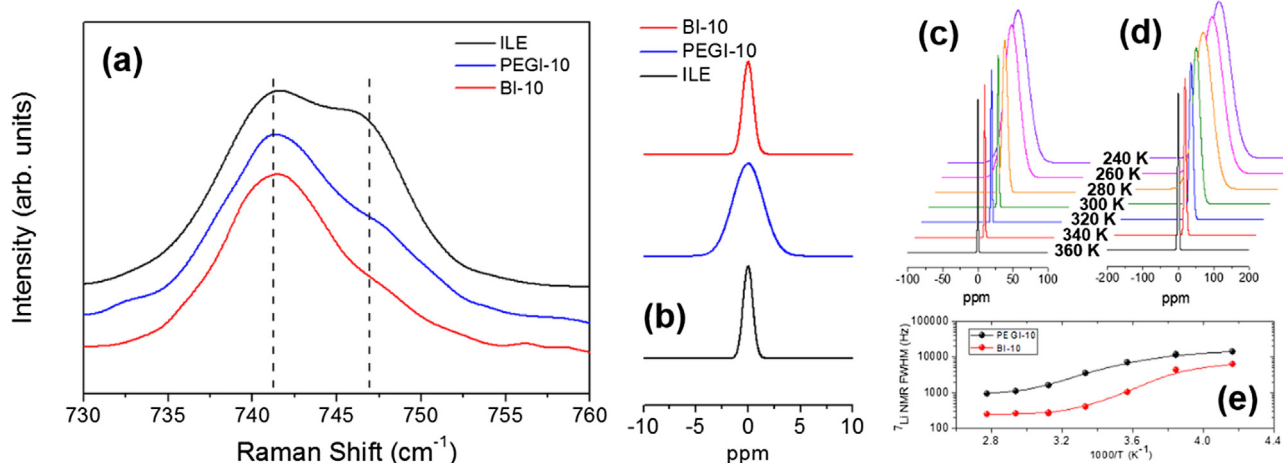


Fig. 6. (a) Raman spectra of ILE, PEGI-10, and BI-10, (b) room temperature solid-state ^7Li NMR spectra, (c) temperature-dependent solid-state ^7Li NMR spectra for BI-10, (d) temperature-dependent solid-state ^7Li NMR spectra for PEGI-10, and (e) solid-state ^7Li NMR spectra linewidth summary for PEGI-10 and BI-10.

boron moieties in the gel state could yield favorable lithium ion transport and thus contribute to the larger ionic conductivity of the materials.

3.4. Lithium-ion battery performance

Subsequently, to evaluate the electrochemical performance of ionogels, LiFePO₄/lithium cells were assembled with prepared ionogels and galvanostatic charge/discharge measurements were conducted in the voltage range of 2.5–4.2 V. LiFePO₄ cathode was selected as active material due to its compatibility with ionogel electrolyte systems and thermal stability [11]. Because the ionogel revealed a sufficient dimensional stability to separate the electrodes in the cells, we were able to exclude the use of separator film. In the cycling response examination of the cell at various C rates (Fig. 7), the cells were charged at a constant current density of 0.1C. As shown, at the initial cycling of 0.05C, the discharge capacity was approximately 142 mAh g⁻¹ for the cell made with BI-10, which was almost similar capacity value to that of the cell made with PEGI-10 gel. However, it was found that the cell with BI-10 exhibited better rate capability with the increase in current density and cycle number. The capacity retention was estimated to be about 47.9% when the current density was increased from 0.1C to 0.5C, whereas the cell with PEG gel maintained capacity retention of 31.6% at a current density of 0.5C. It has been generally reported that a decrease in discharge capacity at a high current rate is primarily coupled with the limited diffusion rate of lithium ion at the electrolyte-electrode interface due to the ionogel quasi-solid state increasing the cell polarization [11]. Moreover from the above observations, we speculated that the increase in rate performance for the BI-10 ionogel could be correlated with the ability of anion-trapping boron moieties to facilitate the lithium ion transport.

The cycling performance of the cells further evaluated through recording of 50 cycles at a constant charging/discharging current rate of 0.1C. In the discharge capacity and Coulombic efficiency diagrams at various cycle number (Fig. 8a), the cell including ionogel with boron moieties exhibited a relatively stable cycling behavior and delivered larger discharge capacity values compared with those of cell with PEGI-10 gel. After cycling, a retained discharge capacity value as large as 130.5 mAh g⁻¹ was achieved, corresponding to 91.5% of its largest discharge capacity. No significant change in capacity values was observed during this prolonged cycles, demonstrating that BI-10 was electrochemically stable and continued to function without appreciable degradation. Moreover, the Coulombic efficiency of the system was found to be stabilized approximately 98% with the exception of initial cycles, verifying that the conversion process between LiFePO₄ and FePO₄ were highly reversible in the cells. Therefore, it can be said that the improved rate capability and cycling stability reflect the

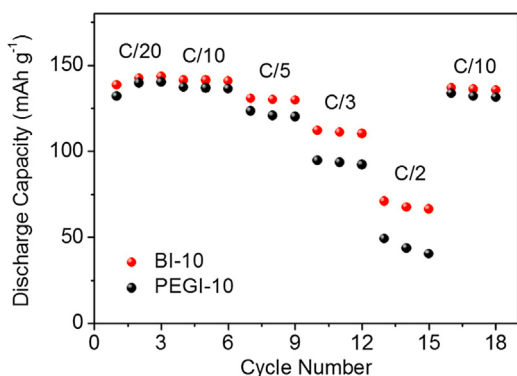


Fig. 7. Rate performances of BI-10 and PEGI-10 ionogel electrolytes.

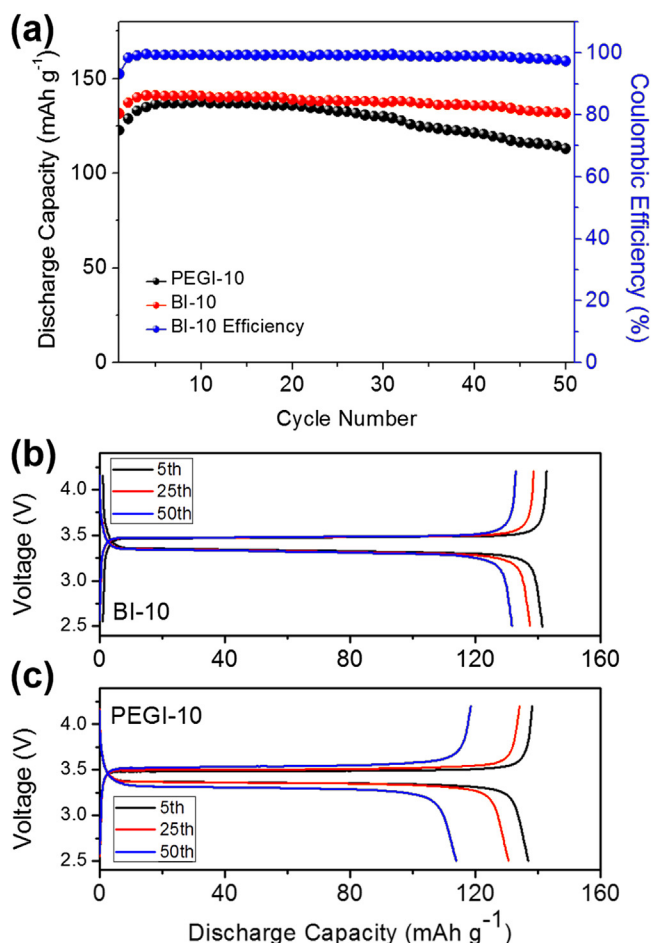


Fig. 8. (a) Cyclability of BI-10 and PEGI-10 ionogel electrolytes, representative charge-discharge profiles at selected cycle numbers for (b) BI-10 and (c) PEGI-10.

advantageous boron-based ionogel material to effectively suppress the increase in polarization and loss of capacity, all attributed to the increase in lithium mobility and anion dissociation character of the anion-trapping boronic crosslinker.

4. Conclusions

A new boron-containing ionogel was fabricated to improve the lithium mobility in solidified ionic liquid matrices. The boronic ionogels exhibited high thermal stability, electrochemical stability, and unique ion conduction behavior. Through the introduction of boronic crosslinker, the lithium mobility was greatly enhanced compared with the ionogels fabricated without the boronic crosslinker, giving promising Li-ion battery performance with regards to both rate and cyclability at elevated temperatures.

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